CS6230 CAD for VLSI: RTL Microarchitecture and Verification Report

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1 Architecture overview and division of work (Spec 5.2)

mkQRDecoder decodes ISO/IEC 18004 QR symbols of all 40 versions (21×21 to 177×177 modules), all four error-correction levels, all eight masks, and Numeric, Alphanumeric and Byte payloads. The design is four streaming stages joined by latency-insensitive FIFOs: **(1) mkQRFrontEnd** reads the two 15-bit format copies (nearest-codeword BCH(15,5) search, 2-stage pipeline, corrects up to 3 bit errors), walks the symbol in two-column zigzag order skipping function patterns, unmasks and packs 8-bit codewords; **(2) mkQRBlocks** stores up to 3,706 codewords (1-cycle store pipeline), de-interleaves them into up to 81 Reed–Solomon blocks and applies the corrections; **(3) mkQRRS / QRGF** perform GF(2^8) arithmetic (polynomial 0x11D), 30-syndrome evaluation, a 2-stage Berlekamp–Massey solver and Chien search with Forney magnitudes, and flag uncorrectable blocks; **(4) mkQRPayload** keeps a 64-bit bit-buffer with single-cycle variable-width extraction, parses mode and character-count fields, checks numeric and alphanumeric ranges and emits the message bytes.

Division of work. To satisfy Spec 5.2 (the author of a unit's reference model does not design its RTL), the work was split between the two team members:

Team member	RTL designed	Reference models and verification written	Synthesis and layout
Kishore (EE26E016)	Backend and GF(2^8): QRBlocks.bsv, QRRS.bsv, QRGF.bsv	Unit Bluesim benches TbGF, TbGFVectors, TbSyndrome, TbBM, TbRS, TbBlocks; 240-symbol end-to-end hardware sweeps (verification/sweep/) that test the Frontend, Payload and Top RTL	All official Design Compiler runs (Table 5) and the ICC PlaceOpt runs (Table 6)
Rishi (EE25S082)	Frontend, Payload and Top: QRFrontEnd.bsv, QRPayload.bsv, QRDecoder.bsv, QRInternalTypes.bsv	Independent Python golden models (verification/model/) and the 76-test pytest suite used on the Backend and GF RTL; TbCornerCases, TbPayloadModes, TbVersion40, TbVersion40Mask3	Four-bank (R5) sub-nanosecond synthesis exploration

Table 1: Who designed and who verified each unit.

2 Microarchitectural optimisations across the five iterations

Five synthesis-driven iterations took the clock from 5.0 ns to 1.2 ns (4.17× faster, 200 → 833.3 MHz) with f2f logic depth falling from 158 to 16 levels; every one met timing with zero violations after synthesis and the best design also after placement.

- **Multi-stage Berlekamp–Massey and clock gating (iterations 1→3).** The single-cycle BM loop was split first into three phases (`-D QR_R2_BM`) and then a 2-stage pipeline (`-D QR_R4_BM`); clock gating (`compile_ultra -gate_clock`) and a 1-cycle store pipeline (`-D QR_STORE_PIPE`) were added. At the same 5.0 ns period this cut power by 54.4 % (9.9178 → 4.5249 mW) and design area by 34.5 %; the clock then closed at 2.5 ns.
- **Divider-free arithmetic and a single-cycle bit-buffer (iterations 4→5).** The integer dividers (/3, /45, /100) in the front end and payload were replaced by exact reciprocal multiply-shifts (×171 9, ×91 12, ×41 12); the BCH search and zigzag updates were pipelined and a single-cycle `readBits` was added. Logic depth fell from 104 to 16 levels and the clock closed at 2.0 ns and

then 1.2 ns.

- **Four-bank store (R5, explored, not selected).** A four-bank codeword store with bit-matrix GF multipliers closed DC at 0.70 and 0.55 ns, but none of its twelve placement runs was clean (1–14 violating paths), so the r4h12 design was kept.

3 Verification methodology and results

Verification has two tiers in `verification/`: (1) a Python golden-model suite (8 files, 76 tests) that also cross-checks all 40 versions against the independent *segno* generator, and (2) Bluesim hardware benches, from exhaustive unit vectors to full-decoder sweeps. One command, `make test` (or `make full`), runs everything.

Check	Stimulus	Result
Python reference models (8 files)	GF, RS, blocks, front end, payload, full decode, protocol; all 40 versions vs segno	76 / 76 tests pass
Unit Bluesim benches (Kishore)	TbGF, TbGFVectors, TbSyndrome, TbBM, TbRS, TbBlocks	all pass
Full-decoder hardware sweeps (two sets)	120 cases each: 30 clean, 60 at the correction limit, 30 beyond it (fail flag expected)	120/120 and 120/120, 0 mismatches
Independent cross-check	zxing-cpp decoder on the generated symbols	90 / 90 agree
Corner cases (Rishi)	format-bit errors, tie-breaker, invalid size, uncorrectable format, …	10 / 10 pass
Payload modes (Rishi)	Numeric, Alphanumeric, Byte, Terminator, mixed	5 / 5 pass
Version 40, mask 3 (Rishi)	largest symbol (177×177), 3,706 codewords	decoded in 38,074 cycles (45.69 µs at 1.2 ns)
Final tree and clean rebuild	make full on the submitted tree; fresh copy of the handout with our bsv/	all steps passed (10 min 7 s); sample symbol in 563 cycles

Table 2: Verification summary. Logs are in verification/results/.

4 Conclusion

The decoder is functionally complete for all QR versions, error-correction levels, masks and payload modes, and is verified at unit, subsystem and full-decoder level with independent reference models. The best configuration closes timing at 1.2 ns (see pd_report.pdf). Known limitation: the physical flow stops at placement; no routing, DRC or LVS was run.